

Peer Skill Exchange & Mentorship Network

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—>**Architecture Selection:** We chose Layered Architecture for the Peer Skill Exchange & Mentorship Network system.

→**Architectural Choice:** The system is designed using a layered architecture, where the presentation layer handles the member interface, the application/service layer manages skills, mentors, sessions and time credits, and the data layer handles the Peer Skill Exchange database.

-Reason 1: Separation of responsibilities makes the system easier to develop and maintain. For example, changes to session scheduling or mentor-management logic can be made in the service layer without changing the user interface.

-Reason 2: The architecture suits the different functions of the system because members can search for skills, find mentors, schedule sessions and complete sessions through the same application, while database operations remain separated from user-facing functions.

Security Advantage: The layered structure allows authentication and authorization checks to be handled before sensitive service or database operations are performed. This helps prevent direct access to member, session and time-credit data.

Performance Benefit: Separating the service and data layers allows database queries to be optimized independently. Frequently requested information such as available skills or mentors can also be cached at the service layer, reducing repeated database access and improving response time.